

Crediting-Cap Design for an Index-Linked Lifetime-Income Product

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This repository is a research implementation of a generic Australian index-linked guaranteed lifetime-income product. It connects contractual cashflows, market-consistent valuation, policyholder behaviour and annual management action in one reproducible portfolio workflow. The central research question is whether an insurer can use an annually reset **crediting cap** to improve the new-business CSM proxy while controlling market, longevity and behaviour risk.

The product, mortality and behaviour bases are illustrative. The reported CSM, risk and capital quantities are research proxies, not recognised IFRS 17 amounts, APRA capital, customer illustrations or financial advice.

1. Introduction

The case study starts with a single-premium Growth phase. The policyholder can choose at an eligible policy anniversary when to enter the Income phase. At that point a fixed nominal lifetime-income amount is locked in. The account continues to receive annually protected reference-fund performance, but there is no income ratchet in the base design.

Products of this kind give customers meaningful timing and liquidity choices, while exposing the insurer to interacting longevity, lapse, withdrawal, interest-rate and option-cost risks. Digital advice and optimisation tools may also make value-sensitive behaviour more relevant than a purely static lapse assumption would suggest. The repository therefore evaluates two deliberately different policyholder models:

- transparent statistical dynamic functions, including moneyness and realised performance signals; and
- a fitted LSMC policy that maximises policyholder value over the admissible Growth and Income actions.

The customer receives the annual simple credit

$$g_y = \min\left(\max\left(R_y^{\text{fund}}, 0\right), C_y\right),$$

where R_y^{fund} is the complete annual reference-fund return and C_y is the Maximum Return announced by the insurer. In this repository, “crediting rate” in workflow names normally means this annual cap, not a guaranteed flat interest rate. The contractual case-study cap is 6%, with a guaranteed minimum cap of 0.25%; alternative caps are design scenarios.

Research problem

The cap affects the account value, fee base, guarantee moneyness, customer actions and the price of the insurer's call spread. Its effect is phase dependent:

Phase	Typical customer channel	Typical insurer channel
Growth	A higher cap increases the potential value of waiting and can raise the income base at election	More fees and later election may help, but the hedge is more expensive
Income	A higher account value does not raise locked income without a ratchet; withdrawal can become the only way to realise gains	Claims may fall, while lapse, longevity exposure, fee duration and hedge cost can move in opposing directions

There is therefore no universally optimal high or low cap. For the current management question, C_y is chosen from a predeclared admissible grid using information available at each decision time. The contractual flexibility is valued today with the primary capital-adjusted objective $CSM^{\text{proxy}} - 0.06K_{\text{MLL}}$ and compared with the best fixed cap on the same complete risk-neutral sample. CSM/MLL remains a supplementary capital-efficiency diagnostic. This is not a deployment or OOS exercise.

Management discretion may also matter to fulfilment-cashflow and service assessments where it is substantive and recognised by the applicable accounting policy. The model only estimates cashflows under an assumed rule; it does not establish IFRS 17 recognition or a group-level CSM.

2. Product from the policyholder's perspective

The generic product is designed to retain the economically important features of comparable lifetime-income contracts without reproducing a particular current insurer offer.

Feature	Case-study rule
Premium	Single premium in AUD
Reference fund	30% Global Equity and 70% rolling five-year nominal Australian government-bond proxy
Rebalancing	Monthly, before the nonlinear annual credit is applied
Annual protection	Negative reference-fund returns credit 0%; positive returns are capped
Growth	No withdrawals; annual irreversible choice to wait or start income after the first full year
Income	Fixed monthly lifetime amount, paid in arrears; no base-case ratchet
Income withdrawals	Contractual excess/partial withdrawal or full surrender, with account and future-income consequences
Automatic start	First anniversary after the primary life reaches age 100
Death and spouse	Single- or joint-life treatment with the elected spouse-death continuation rule

The combined monthly fund return is

$$R_m^{\text{fund}} = 0.30R_m^{\text{global}} + 0.70R_m^{\text{bond}}.$$

Equity and bond returns are combined first; only then is the annual floor/cap payoff applied. A higher cap during Growth can benefit the policyholder through a larger account and election-date income base. After Income starts, later positive credits do not increase the locked payment in the base design. Ratchet variants can be offered in practice, but commonly exchange that upside for a lower initial conversion rate. Under illustrative product comparisons it can take roughly 8–15 years for the ratcheted income to catch the initially higher fixed payment; this range is design-dependent and is not modelled as a universal market fact here.

Scheduled income first uses the account value. When that value is exhausted, the insurer funds the covered shortfall for as long as an eligible life survives. Growth-to-Income election, voluntary Income actions, spouse coverage and the exact anniversary order are described in [the product design](#).

3. Product from the insurer's perspective

The account is an administrative customer benefit account; it is not assumed to be invested directly in the reference fund. Customer money is instead held in a money-market backing account and its pathwise return belongs to the insurer. Customer index participation is manufactured separately with a capital-market bull call spread:

$$\min(\max(R, 0), C) = \max(R, 0) - \max(R - C, 0).$$

The insurer buys the lower call and sells the cap call **to the capital market**, not to the customer. Raising the cap reduces the value received for the sold upper call and therefore increases net hedge cost. The standard case uses the complete sold call spread. A research alternative in which the upper call is not sold must be labelled explicitly; only there can performance above the customer cap become retained hedge income.

The standard profitability objective is the signed new-business CSM proxy

$$\text{CSM}^{\text{proxy}} = \text{PV}_{\text{fees}} + \text{PV}_{\text{other}} - \text{PV}_{\text{claims}} - \text{PV}_{\text{costs}}.$$

Leg	Main modelled components
Fee Income	Product fee and lifetime-income premium
Other Income	Money-market backing return, MVA/other retained margins and, only where applicable, retained hedge gain
Claims	Guarantee shortfalls and separately identified insurer-funded benefits
Costs	Acquisition and maintenance expense plus the complete option package: fair value, purchase markup and hedge-reference management fee

Customer payments funded by the account value are investment-component cashflows and are not deducted a second time as insurer claims. Every material report must reconcile the CSM proxy to these four legs. See [crediting rate, profitability and risk](#) for the economic channels and accounting boundary.

Money-market backing can provide a natural partial offset to movements in the discounting of customer cashflows, but it is neither a perfect interest-rate hedge nor the return credited to the policyholder.

4. Modelling approach

Portfolio and assumptions

The full illustrative portfolio contains 48 model points. A four-point proxy is provided for faster development. Customer-LSMC runs deliberately require the one-point proxy. The Time-0 Management-LSMC calculation in Section 7 also hard-requires that same single modelpoint; no 4- or 48-point aggregation is used in that calculation. A one-modelpoint output is a method/design sensitivity, not portfolio evidence. A model point represents an insured person; scalar portfolio results use `contract_weight`, not an implicit count of CSV rows. The [input reference](#) documents the portable files under `input_data/`.

The base expense assumptions include acquisition and maintenance costs. The insurer is also assumed to pay 0.50% of the fair option-package value as a purchase markup and 0.30% per year of hedge-reference notional as a management fee. The first is a relative markup on option value, not 50 basis points of notional.

Economic scenario generator

Regular market-consistent valuations use correlated Heston–Hull–White dynamics under the risk-neutral measure. The Hull–White curve fits the Australian zero curve at time zero; equities have stochastic variance and configured equity/equity and rate/equity correlations. The Australian curve is the only live market series. No volatility surface, credit-spread curve or separate calibration history is required by the baseline.

All operative Q readers load exact, validated monthly paths and discount factors from `results/cache/q_market_paths`. Conditional-MC prices for each new annual call spread are loaded from the path-congruent `results/cache/q_hedge_prices`. Only the dedicated precompute runner may write those directories. The risk orchestrator and the installed crediting-cap optimisation commands validate existing entries and ask that runner to create only missing exact entries before their read-only children begin. A changed cap grid or equity allocation creates a distinct hedge-cache identity but can reuse the same exact market cache. An approximate cache match and an implicit Black–Scholes substitution are prohibited; `moment_matched_bs` is available only as an explicit, labelled proxy choice.

Simplified real-world projections use Black–Scholes–Hull–White under the physical measure with the supplied equity risk premia, no bond term premium and the same rate dynamics as Q. They are proxy projections, not forecasts. The full conventions are in [modelling methodology](#).

Mortality and behaviour

The mortality basis is an illustrative Gompertz–Makeham table with annual improvement, reconciled monthly decrement probabilities and explicit joint-life states. It is not calibrated insured-life experience. See [mortality modelling](#).

Dynamic behaviour uses duration baselines and bounded hazard/link functions. Moneyiness, premium size, MVA and the gap between gross reference performance and credited performance can alter Income take-up, lapse and excess withdrawal. Customer LSMC instead treats the contract as one ordered, swing-option-like problem and maximises the time-zero customer objective

$$\mathbb{E}_0 \left[\sum_t P^{\text{AU}}(0, t) CF_t^{\text{customer}} \right],$$

using the deterministic discount factors implied by today's Australian zero curve. The mortality-free fit uses one exact Q-market sample and deploys the fitted V11 policy directly, without an independent policy-selection sample or replacement by a fixed rule. Complete-path fold cross-fitting remains inside the continuation-value estimator; it estimates conditional expectations. At annual Growth decisions the customer chooses `WAIT` or `START_NORMAL_INCOME`; at annual Income decisions the choice is normal income for the next period or `FULL_SURRENDER`. Partial withdrawal and mortality are absent from the customer objective. At the finite projection horizon the terminal payoff is the post-fee account-value closeout. Customer-LSMC runs require exactly one model point. See [policyholder behaviour](#).

Crediting-cap control and core scripts

The insurer chooses the next cap after old-year crediting, fees, mortality and eligible Income election, but before the new hedge is purchased. Optimisation uses only pre-action state. The Dynamic-customer management valuation uses one complete Q sample both to fit conditional expectations and to determine today's risk-neutral capital-adjusted CSM; it has no OOS test, forward roll or deployment gate. CSM/MLL is reported alongside the primary AUD objective. This is distinct from the separate Policyholder-LSMC/Stackelberg research route.

The small set of scripts that defines the operative research workflow is:

Script	Role
<code>run_portfolio_risk_analysis.py</code>	Recommended end-to-end fixed-cap, behaviour and optional shock-and-revalue orchestrator; it prepares exact caches before invoking readers
<code>run_crediting_rate_capital_analysis.py</code>	Dynamic-only fixed-cap mortality/longevity/lapse capital and capital-adjusted profitability orchestrator; Policyholder LSMC is hard-blocked
<code>run_crediting_rate_optimisation.py</code>	Console-command orchestrator for both optimisation families; prepares only missing exact caches, then starts a strict reader
<code>optimize_crediting_rate_dynamic_behaviour_alt.py</code>	Strict cache-reader for the one-modelpoint, same-sample Time-0 capital-adjusted CSM value of annual cap flexibility under statistical Dynamic Policyholder behaviour; CSM/MLL is a secondary diagnostic
<code>optimize_crediting_rate_bellman.py</code>	Strict cache-reader LSMC-policyholder entry point; delegates to the combined Stackelberg implementation
<code>precompute_q_market_and_hedge_cache.py</code>	Sole authorised writer of exact Q-market and conditional-MC hedge caches
<code>run_portfolio_valuation.py</code>	Read-only dynamic-behaviour portfolio valuation
<code>run_portfolio_valuation_ls_mc.py</code>	Read-only one-modelpoint customer-LSMC valuation on one exact Q sample with direct V11 deployment

The optimisation details are in [crediting-rate optimisation](#); the orchestration and output contract are in [portfolio valuation and risk workflows](#). A complete active-runner inventory is in the [script reference](#).

5. Optimal versus dynamic policyholder behaviour

The fixed-cap comparison values two complete policies on the same exact cached Q sample:

1. statistical dynamic Income election plus dynamic Income lapse/withdrawal;
2. the directly deployed V11 customer-LSMC policy.

The comparison reports CSM and its components, guarantee claims, election timing, surrender diagnostics and risk sensitivities. V11 is the direct single-sample policy once the structural regression checks succeed. Its internal complete-path folds estimate continuation values.

The one-modelpoint diagnostic uses 20,000 common Heston–Hull–White paths for `ALT4-01` and the discrete cap grid 0.25%, 0.5%, 1%, 2%, 4%, 6%, 8% and 12%. V11 is valid and deployed directly in every displayed cell. Its customer fit and the Dynamic arm use the same exact market sample. The base market model provides the complete grid, so no alternative fixed-equity-volatility or lower-rate-volatility sensitivity is used.

This is an illustrative result for one representative contract, not evidence about a diversified portfolio. The full sample identities, method flags and outputs are in the [curated source table](#).

The optimal decisions change in annual steps rather than along a smooth cap response. Mean V11 Income Election occurs in year 1 for caps from 0.25% through 2%, around year 3 at 4%, again around year 1 at 6%, and around year 4 at 8% and 12%. This non-monotone pattern reflects the pathwise trade-off between waiting in Growth and starting normal Income at the permitted annual decision dates; it must not be interpolated between cap scenarios. Full Surrender is zero to displayed precision apart from negligible path mass at 12%. For this model point, the value difference is therefore driven mainly by Income-Election timing and the resulting normal-income cashflows rather than by surrender.

The customer-value comparison below puts both behaviour models on the same pathwise Q valuation basis and uses the same contractual benefit definition. The difference is therefore the paired increase in Policyholder-benefit PV from the V11 customer rule relative to Dynamic behaviour.

Cap	Dynamic customer benefit PV (AUD)	V11 customer benefit PV (AUD)	V11 increase (AUD)	V11 increase
0.25%	246,411.57	301,761.23	55,349.65	22.46%
0.5%	248,964.57	302,608.32	53,643.75	21.55%
1%	254,137.13	304,296.09	50,158.96	19.74%
2%	264,492.22	307,636.96	43,144.75	16.31%
4%	284,955.98	306,866.85	21,910.87	7.69%
6%	303,121.48	319,908.63	16,787.15	5.54%
8%	317,300.20	326,555.95	9,255.76	2.92%
12%	332,245.38	337,256.46	5,011.08	1.51%

The V11 customer rule increases the customer-benefit PV at every displayed cap. The uplift is largest at low caps, where earlier Income Election avoids much of the value loss under Dynamic behaviour, and narrows from 22.46% at 0.25% to 1.51% at 12% as the two realised benefit profiles converge. This is a customer-benefit comparison, not the insurer CSM effect or the separate optimisation objective used to fit V11.

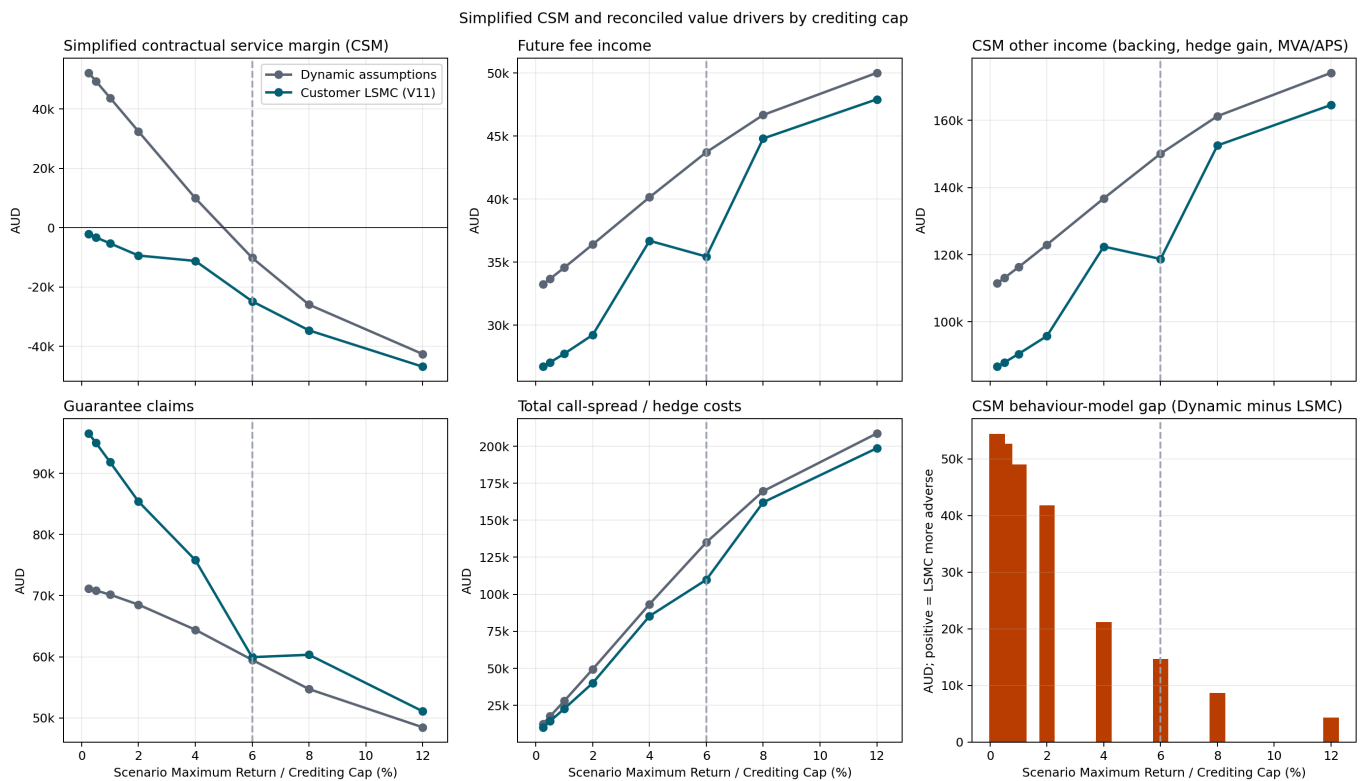
6. Effect of the crediting rate

A controlled cap study varies only *C*, keeps common random numbers and revalues both behaviour models. The key outputs are:

- CSM proxy and the four-leg reconciliation by cap;
- option fair value, markup, management fee and money-market income by cap;
- guarantee claims, election timing and voluntary-action rates by cap; and
- shock-and-revalue differences for market and non-market stresses.

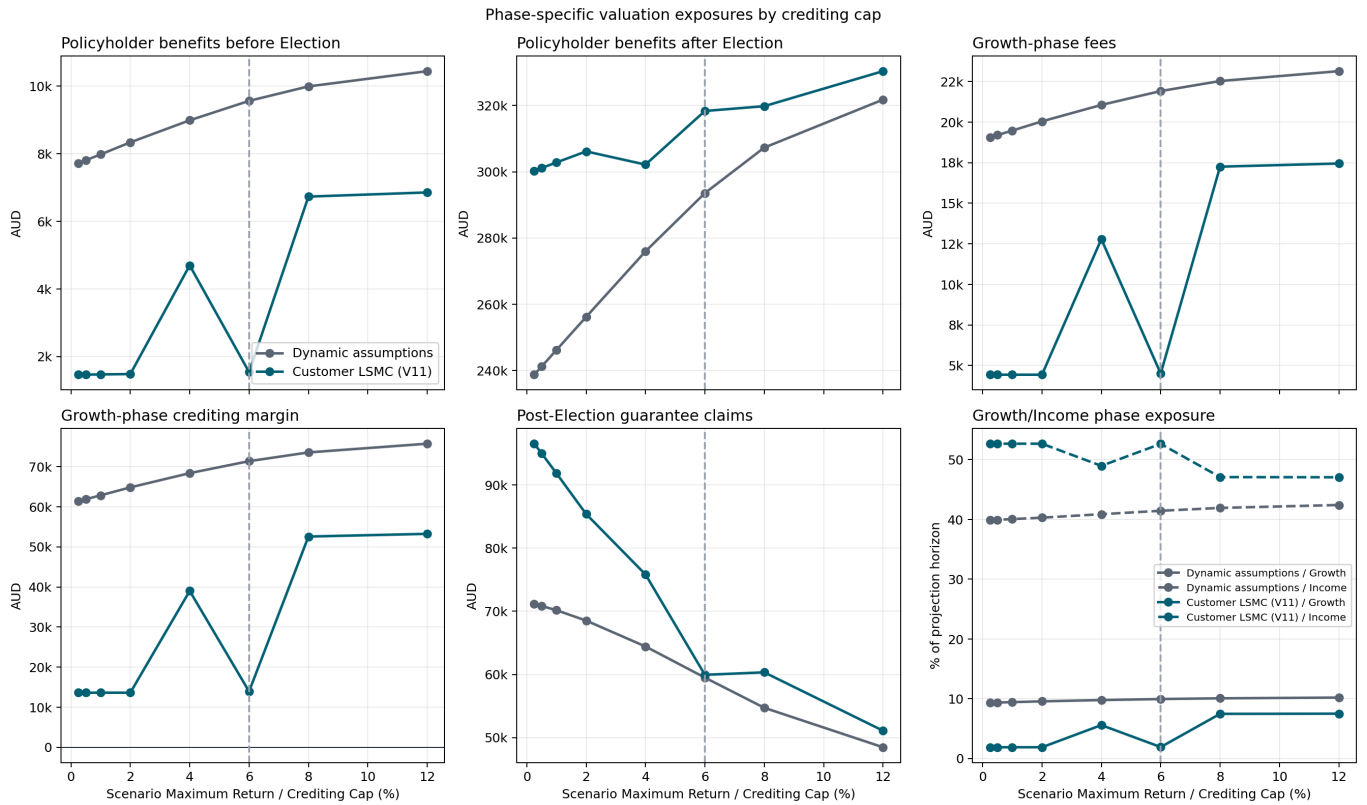
A higher cap is expected to increase hedge cost, but the net CSM and risk effects need not be monotone because account value, fee duration, claims and behaviour all respond. Across the displayed one-modelpoint cap grid, the observed values are:

Cap	Dynamic CSM (AUD)	Direct V11 CSM (AUD)	Customer optionality uplift (AUD)	Mean V11 Income-start year
0.25%	52,128.84	-2,225.25	0.00	1.0000
0.5%	49,350.86	-3,274.68	0.00	1.0000
1%	43,713.58	-5,316.92	0.00	1.0000
2%	32,394.84	-9,412.17	0.00	1.0000
4%	9,926.33	-11,222.41	34.06	2.9819
6%	-10,170.40	-24,811.80	3,684.45	1.0171
8%	-25,942.61	-34,606.39	13,426.51	4.0001
12%	-42,571.89	-46,840.27	66,949.93	4.0146



Guarantee claims generally decline as the cap rises, while call-spread cost grows from AUD 10,137 at 0.25% to AUD 198,614 at 12%; V11 CSM consequently falls from AUD -2,225 to AUD -46,840.

The optionality uplift is the increase in the deterministic-time-zero-curve customer objective relative to the best fixed START-plus-CONTINUE reference. It is neither an insurer CSM increment nor a separate-sample performance estimate. These are discrete, base-only scenarios for ALT4-01, not interpolated break-even estimates or portfolio-level evidence.



The discrete V11 start-year changes at 4%, 6% and 8% shift value and exposure between Growth and Income.

The two figures in this section were rendered data-only from the unified, validated result table. Figure hashes, cap-cell identities and exact source hashes are recorded in the [current figure provenance](#).

Sections 5 and 6 are a separate Customer-LSMC diagnostic and are not inputs to the Management-LSMC valuation below, which uses statistical Dynamic customers and current-curve Time-0 cashflows.

7. Value of Optimal Management Decisions for the Crediting Rate

The question here is not how to deploy a cap strategy. It is the value today of the insurer's contractual right to reset the annual crediting cap in future, given the information available at each future anniversary. Future cashflows are risk-neutral expected values discounted back to Time 0 using the current Australian curve.

The illustrative Time-0 valuation uses exactly one modelpoint (ALT4-01), 4,200 common Heston-Hull-White Q paths with market seed 2026, and exact path-congruent market and hedge caches. Policyholders follow the statistical Dynamic behaviour model, including its moneyness-sensitive lapse and election response. No Customer LSMC is fitted or called.

The fixed caps and the management LSMC are evaluated on the same complete Q sample. There is deliberately no reserved path subset, different OOS seed, forward roll, strategy replay, bootstrap acceptance test or deployment rule. The output is a Time-0 valuation, not an estimate of live strategy performance.

For each complete management candidate π , the primary objective is the AUD capital-adjusted value

$$J_{\text{adj}}(\pi) = \text{CSM}^{\text{proxy}}(\pi) - 0.06K_{\text{MLL}}(\pi).$$

Here K_{MLL} combines mortality, longevity and the largest adverse lapse amount, including the separately disclosed 40% positive-CSM mass-lapse proxy. It remains a partial research life-risk capital measure, not total SCR, APRA capital or an IFRS 17 Risk Adjustment. The 6% factor is a one-year research capital hurdle, not a projected Risk Margin. The lifetime efficiency ratio $\text{CSM}/K_{\text{MLL}}$ is reported as a secondary diagnostic.

MLL and its stress maxima are not additive annual Bellman rewards. Management LSMC therefore fits a predeclared class of 21 additive Base/Stress support objectives,

$$J_{\alpha,q} = (1 - \alpha) \text{CSM}_{\text{base}} + \alpha \sum_s q_s \text{CSM}_s, \quad \alpha \in \{0, 0.25, 0.50, 0.75, 1\},$$

plus one conditional-ratio heuristic. The exact aggregate J_{adj} —not a support objective—ranks the candidate set. Each fitted payload difference is anchored to the directly projected best fixed cap, and best fixed remains an explicit zero-flexibility-value comparator.

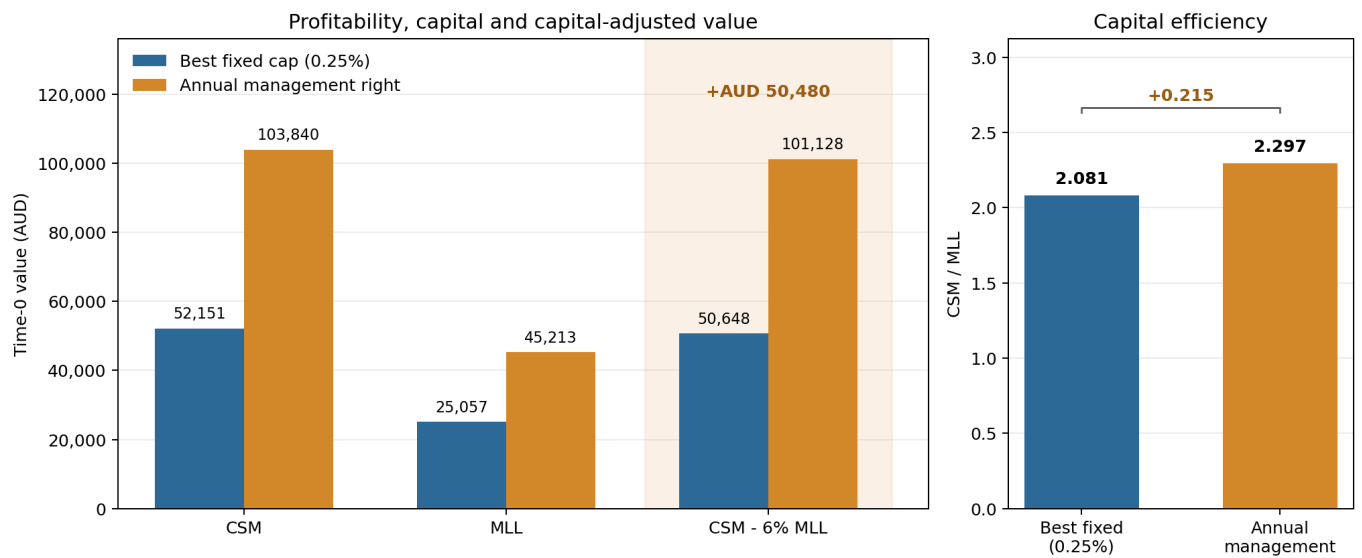
The stored candidate table contains all 22 fitted candidates. Earlier ratio-only reporting selected the balanced four-stress candidate with $\alpha = 0.50$; that result is retained only as a historical objective sensitivity. Applying the current capital-adjusted objective to the complete candidate table selects the pure `base_csm` candidate. This deterministic re-ranking required neither a new market projection nor a regression refit.

Time-0 alternative	CSM (AUD)	MLL (AUD)	CSM – 6% MLL (AUD)	CSM / MLL
Best fixed cap: 0.25%	52,151.27	25,057.48	50,647.82	2.08127
Annual adjustment right: <code>base_csm</code>	103,840.38	45,213.13	101,127.59	2.29669
Flexible minus fixed	+51,689.11	+20,155.65	+50,479.77	+0.21542

The fitted candidate's first Time-0 action is also 0.25%. Its additional value comes from the right to make later state-dependent resets, not from choosing a different initial cap. No future deployment schedule is exported.

Optimal annual crediting-rate management improves value and efficiency

Direct visualisation of the Section 7 comparison table



This single graphic is a direct visualisation of the table above; no additional fixed-cap curves or risk-module charts are included. It shows that the primary capital-adjusted value rises by AUD 50,479.77 and the supplementary CSM/MLL ratio by 0.21542. Absolute MLL increases by AUD 20,155.65 because CSM increases by AUD 51,689.11, but MLL per unit of CSM falls from 48.05% to 43.54%. The observed benefit is therefore higher capital-adjusted value together with better partial life-risk efficiency, not a reduction in absolute capital.

This evidence has important boundaries. It uses exactly one illustrative modelpoint, and MLL covers mortality, longevity and lapse only. The 40% positive-CSM mass-lapse proxy can materially shape both the AUD objective and the ratio. The result is the best member of the declared fitted policy class, not a global optimum over all management rules. It is in-sample by design, and regression fitting, anchoring, path count and stress calibration remain model risk. These limitations are consistent with a Time-0 valuation claim but rule out interpreting the result as tested strategy performance.

All calculations in this section currently use statistical Dynamic Policyholder behaviour only. Policyholder behaviour optimised by a separate Customer LSMC has not been included in the Section 7 values. Repeating the Time-0 management-flexibility analysis with an LSMC Policyholder response is an interesting extension for future research, because optimal customer decisions could change both CSM and the mortality, longevity and lapse capital profile.

Machine-readable settings, the full fixed-cap grid, the complete fitted candidate table and regression diagnostics are stored with the [supporting figure data](#).

Quickstart

Python 3.10 or newer is required. The supported portable setup is a source checkout with an editable install, because `input_data/` remains a repository-relative data tree rather than wheel package data. From the cloned repository root:

```
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -e ".[test]"
python -m pytest
```

The recommended end-to-end fixed-cap workflow is the risk orchestrator. Runs that include customer LSMC default to and require exactly one model point. This example creates or validates every exact required cache before the read-only valuation children start:

```
portfolio-risk-analysis --model-points
    input_data/model_points_policyholders/model_points_policyholders_1_point_proxy.csv --crediting-
    rates 0.25% 1% 6% 12% --require-market-cache --require-hedge-cache
```

Add the preselected market/longevity shock grid explicitly:

```
portfolio-risk-analysis --model-points
    input_data/model_points_policyholders/model_points_policyholders_1_point_proxy.csv --crediting-
    rates 0.25% 1% 6% 12% --stress-analysis --stress-scenarios interest_up interest_down longevity --
    require-market-cache --require-hedge-cache
```

The two installed optimisation commands provide the same prepare-then-read boundary. They derive the horizon, required path counts, seeds and complete cap grid from the optimiser arguments, invoke the sole authorised cache writer for missing exact entries, and only then start the strict reader. The Dynamic route prepares one complete Time-0 Q sample and rejects anything other than one modelpoint before cache preparation:

```
optimise-crediting-dynamic --model-points
    input_data/model_points_policyholders/model_points_policyholders_1_point_proxy.csv --n-paths 4200
    --seed 2026 --require-market-cache --require-hedge-cache
optimise-crediting-lsmc --model-points
    input_data/model_points_policyholders/model_points_policyholders_1_point_proxy.csv
```

These runs can be computationally and disk intensive. Direct execution of a valuation or optimiser implementation file does not perform cache preparation and requires exact pre-existing caches. If such a reader reports a cache miss, run the installed orchestrated command or use its exact precompute specification rather than changing the seed, horizon, path count, allocation or cap grid to reach an approximately matching entry.

Repository map

Path	Purpose
code/policy_engine/	Reusable product, scenario, projection, valuation, mortality and behaviour engine
code/portfolio_simulations/	Sole cache writer, prepare-then-read orchestrators, strict valuation readers, risk analysis and cap optimisation
code/code_from_papers/	Reference implementations derived from research papers; not the production workflow
input_data/	Versioned portable market, cost, behaviour, allocation, Monte Carlo and model-point inputs
documents/	Current English product, method and workflow documentation
tests/	Unit, reconciliation, regression and orchestration tests
results/cache/	Generated exact Q-market and path-congruent hedge caches; only the precompute runner writes them
results/runs/	Generated run directories and manifests; ignored by Git
results/document_figures/	Reserved for reviewed, small, versioned documentation figures
old/	Historical AGILE-specific material and superseded research artefacts; not an execution surface

Portable default paths are resolved by `repository_paths.py`; cloning the repository does not require editing machine-specific paths. The architectural dependency direction and provenance rules are documented in `engine architecture`.

Selected literature

The references below were selected for their direct connection to this repository rather than as a comprehensive survey. RILAs, variable annuities and GMWB/GLWB contracts are the closest published analogues, but none is identical to the generic case-study product. The accounting and prudential standards define reporting boundaries; they do not validate the repository's CSM or capital proxies.

Index-linked lifetime income and policyholder behaviour

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